Uncertainty-Aware Scientific Claim Synthesis: A Multi-Agent Framework for Contradiction Detection, Gap Formalization, and Confidence-Grounded Literature Review Generation

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Abstract. The exponential growth of scientific publications has created an unprecedented challenge for researchers attempting to synthesize knowledge, identify genuine research gaps, and navigate conflicting findings across literature. Existing automated literature review systems treat papers as flat, unordered collections and generate summaries that lack epistemic grounding, ignore inter-paper contradictions, and produce vague, unverifiable gap claims. This paper presents a novel autonomous multi-agent framework that addresses these fundamental limitations through five core technical contributions: (1) a *Confidence-Graded Claim Extraction and Reliability Scoring* (CGCERS) mechanism that assigns uncertainty estimates to extracted scientific claims based on multi-paper consensus analysis; (2) a *Contradiction Detection and Resolution Agent* (CDRA) that identifies, classifies, and reconciles conflicting assertions using a structured four-dimensional conflict taxonomy; (3) a *Formalized Research Gap Ontology* (FRGO) that produces structured, verifiable gap objects with falsifiability statements; (4) a *Dynamic Scientific Knowledge Graph* (DSKG) built on NetworkX serving as the central intermediate representation; and (5) a *Temporal Knowledge Evolution Tracker* (TKET) for modeling diachronic research trajectories. The system is orchestrated by an autonomy loop controller that iteratively refines outputs based on critic agent feedback. We implement the complete framework using Google Gemini as the primary reasoning engine with a modular architecture supporting provider switching, and deploy a Streamlit-based interactive frontend for visualization and exploration.

Keywords: Multi-Agent Systems · Literature Review Generation · Scientific Knowledge Graphs · Contradiction Detection · Research Gap Identification · Uncertainty Quantification · Large Language Models

1 Introduction

1.1 Motivation

Scientific knowledge is growing at a rate that fundamentally outpaces human capacity to synthesize it. In 2023 alone, over 2.8 million peer-reviewed articles were indexed across major academic databases [1]. A single research domain such as transformer-based natural language processing now encompasses thousands of active papers, with findings that frequently build upon, contradict, or supersede one another. The literature review—the foundational activity of any research endeavor—has become a bottleneck of enormous proportion.

Manual literature review is time-consuming, subjective, and increasingly infeasible at scale. A thorough review of even a moderately sized domain can take months of expert effort, and the resulting document reflects the cognitive limitations of a single reviewer: selective coverage, unconscious bias toward familiar clusters of work, and an inability to systematically detect contradictions spread across hundreds of papers [2].

Automated approaches have attempted to address this problem, but they introduce a different class of failures. Summarization-based systems produce fluent text that lacks epistemic grounding. Pipeline-based agent systems treat synthesis as retrieval followed by concatenation [3]. None of the existing systems ask—and answer—the most important questions a skilled reviewer asks: *How confident should I be in this claim? Which papers disagree with each other, and why? Where exactly is the frontier of knowledge, and what is the most specific gap I can formulate?*

This paper proposes a system architecture that places these three questions at its core.

1.2 Research Motivation Through a Concrete Example

Consider the domain of retrieval-augmented generation (RAG). A researcher entering this domain encounters papers claiming that RAG consistently outperforms parametric-only models on knowledge-intensive tasks, while other papers demonstrate that RAG performance degrades significantly for multi-hop reasoning [6]. Still others argue the comparison is unfair because benchmarks differ. A naive summarization system produces a paragraph that smooths over these contradictions. A skilled human reviewer identifies the disagreement, classifies it as methodological, and formulates a specific gap. Our system is designed to replicate this expert reasoning process autonomously.

1.3 Key Limitations of Prior Work

Existing systems including AutoSurvey [2], LitLLM [3], SurveyAgent [4], and ResearchAgent [5] share the following structural limitations:

1. **Flat Paper Representation:** Papers are stored as embeddings or text chunks without structured relational modeling, making cross-paper contradiction detection effectively impossible.
2. **Ungrounded Claim Generation:** Generated reviews assert claims without any mechanism for tracking how many papers support a claim, whether supporting papers are recent, or whether counter-evidence exists.
3. **Vague Gap Identification:** Gap statements are generated by prompting an LLM to identify "what is missing," producing generic, unverifiable, and untethered claims.
4. **Static, Non-Temporal Synthesis:** Papers published in 2018 and 2024 are treated equivalently, ignoring the temporal evolution of scientific consensus.
5. **Evaluation Inadequacy:** ROUGE and BERTScore measure surface similarity, not synthesis quality, contradiction coverage, or gap accuracy [7].

1.4 Contributions

This paper makes the following novel technical contributions:

- **CGCERS:** A mechanism that extracts atomic scientific claims, maps them across the corpus, and assigns confidence scores based on multi-paper consensus, recency weighting, and counter-evidence presence.

- **CDRA:** A specialized two-stage agent that identifies conflicting claims, classifies conflicts using a four-dimensional taxonomy (methodological, domain-specificity, temporal, definitional), and generates reconciliation statements.
- **FRGO:** A structured representation of research gaps as typed, evidence-grounded objects with explicit falsifiability statements enabling empirical evaluation.
- **DSKG:** A NetworkX-based heterogeneous directed graph replacing flat vector stores, with five node types and eight epistemic edge types enabling graph-based synthesis reasoning.
- **TKET:** A module modeling the diachronic trajectory of research threads, tracking consensus shifts, method emergence, and claim qualification over time.
- **Novel Evaluation Protocol:** Including Claim Coverage Score (CCS), Contradiction Detection F1 (CD-F1), Gap Precision at K (GP@K), Temporal Coherence Score (TCS), and Confidence Calibration Score (CalibScore).

2 Problem Statement

2.1 Formal Problem Definition

Let $\mathcal{P} = \{p_1, p_2, \ldots, p_n\}$ be a corpus of n research papers relevant to a user-specified research topic T. Each paper p_i contains a set of scientific claims $\mathcal{C}_i = \{c_{i,1}, c_{i,2}, \ldots\}$, where each claim c is a propositional assertion about methods, findings, datasets, or theoretical positions.

We define the following sub-problems:

Sub-problem 1 — Confidence-Graded Claim Extraction: Extract the complete claim set $\mathcal{C} = \bigcup_i \mathcal{C}_i$ and for each claim $c \in \mathcal{C}$, compute a confidence score $\sigma(c) \in [0, 1]$ reflecting multi-paper consensus and recency.

Sub-problem 2 — Contradiction Detection: Identify the contradiction set $\mathcal{X} = \{(c_a, c_b) \mid c_a \text{ conflicts with } c_b\}$. For each contradiction, produce a resolution statement r and a conflict class $\kappa \in \{$methodological, domain-specificity, temporal, definitional$\}$.

Sub-problem 3 — Formalized Gap Generation: Generate a structured gap set $\mathcal{G} = \{g_1, g_2, \ldots\}$ where each gap g is a typed object with grounded evidence, specific enough to be empirically evaluated against future publications.

Sub-problem 4 — Structured Review Generation: Synthesize a coherent literature review $\mathcal{R}$ that integrates $\mathcal{C}$, $\mathcal{X}$, and $\mathcal{G}$ with epistemic grounding, thematic organization, and temporal awareness.

2.2 Research Questions

This work addresses four primary research questions:

RQ1: Can a multi-agent system automatically extract scientific claims and assign calibrated confidence scores that correlate with expert assessments?

RQ2: Can automated contradiction detection achieve recall and precision comparable to expert reviewers?

RQ3: Can formalized research gap objects be empirically validated by correlating with papers published after the corpus cutoff date?

RQ4: Does a Knowledge Graph intermediate representation produce measurably more coherent reviews compared to flat vector store baselines?

3 Related Work

3.1 Automated Literature Review Systems

Early automated review systems relied on extractive summarization, selecting and concatenating relevant sentences from source papers [8]. The introduction of transformer-based language models enabled abstractive summarization with systems like ScisummNet producing more fluent summaries [9]. However, fluency does not imply synthesis quality.

More recent systems including AutoSurvey [2] adopt multi-stage LLM pipelines for survey generation. LitLLM [3] introduced a human-in-the-loop approach that improves relevance at the cost of autonomy. SurveyAgent [4] applied ReAct-style reasoning to survey generation, enabling iterative query refinement. These systems represent genuine progress but share the fundamental limitation of treating the paper corpus as a flat retrieval space rather than a structured relational knowledge base.

3.2 Multi-Agent Reasoning Systems

The ReAct framework [10] established the foundation for interleaving reasoning and action in language model agents. Reflexion [11] extended this by introducing self-evaluation and memory for iterative improvement. More recently, LangGraph [12] and CrewAI [13] have provided frameworks for orchestrating multiple specialized agents with defined roles and communication protocols. Our system builds upon these foundations but introduces agent specializations designed specifically for scientific reasoning tasks.

3.3 Scientific Knowledge Graphs

Knowledge graphs have been applied to scientific literature through systems such as ORKG (Open Research Knowledge Graph) [14] and ScholarlyKG [15]. These systems require substantial human annotation effort. Our Dynamic Scientific Knowledge Graph is constructed autonomously from extracted claims, with the novel addition of epistemic edge types (supports, contradicts, extends, qualifies) enabling contradiction detection at the graph level.

3.4 Research Gap Identification

Research gap identification has received limited formal treatment. Most systems generate gap statements through direct LLM prompting or identify gaps as absent topics [16]. Our formalized gap ontology represents, to our knowledge, the first attempt to define research gaps as typed, evidence-grounded, empirically evaluable structured objects with explicit falsifiability statements.

4 Proposed Methodology

4.1 System Overview

The proposed system follows an iterative autonomous loop structured around a central Dynamic Scientific Knowledge Graph (DSKG) as its core intermediate representation. Unlike existing systems that treat synthesis as a linear pipeline, our architecture is fundamentally graph-centric: all agents read from and write to the DSKG, enabling collective reasoning across the entire agent ensemble. Fig. 1 illustrates the high-level execution flow.

4.2 Confidence-Graded Claim Extraction and Reliability Scoring (CGCERS)

Atomic Claim Extraction. The Claim Extractor Agent processes each paper through a structured prompt that instructs the LLM to decompose the paper into atomic propositional claims. An atomic claim is defined as a single, indivisible assertion that can be independently evaluated as true or false within a defined context. Claims are extracted at three levels:

- **Method Claims:** Describing what technique or approach was proposed or used.
- **Result Claims:** Describing empirical outcomes, benchmark scores, or observed phenomena.
- **Theoretical Claims:** Describing proposed explanations, hypotheses, or theoretical positions.

Each extracted claim is stored with its source paper identifier, section location, claim type label, confidence indicators (hedging language such as “suggests” vs. “demonstrates”), subject entities, and condition qualifiers.

Confidence Scoring Formula. For each claim c extracted from paper p_i, the confidence score $\sigma(c)$ is computed as:

$$\sigma(c) = \alpha \cdot \text{Consensus}(c) + \beta \cdot \text{Recency}(c) - \gamma \cdot \text{Contradiction}(c) \tag{1}$$

where:

– Consensus(c) is the proportion of papers in the corpus containing a semantically equivalent or supporting claim, computed via embedding similarity threshold matching against the DSKG.
– Recency(c) is a time-decay weighted score: Recency(c) $= \frac{1}{|S_c|} \sum_{p_j \in S_c} e^{-\lambda(Y_{\text{current}} - Y_j)}$, where Y_j is the publication year and λ is a decay constant.
– Contradiction(c) is the proportion of papers containing claims flagged as conflicting with c by the CDRA.
– α, β, γ are weighting parameters set to $\alpha = 0.4$, $\beta = 0.35$, $\gamma = 0.25$.

The resulting score categorizes each claim as shown in Table 1.

Table 1. Confidence categories derived from the CGCERS score $\sigma(c)$.

Category	**Score Range**	**Interpretation**
High Confidence	$\sigma(c) \geq 0.75$	Well-established, multi-paper support
Moderate Confidence	$0.50 \leq \sigma(c) < 0.75$	Supported but limited evidence
Contested	$0.25 \leq \sigma(c) < 0.50$	Conflicting evidence exists
Disputed	$\sigma(c) < 0.25$	Significant counter-evidence

4.3 Contradiction Detection and Resolution Agent (CDRA)

Two-Stage Detection Architecture. Contradiction detection operates at two levels, balancing computational efficiency with detection accuracy:
Stage 1 — Candidate Pair Identification: For each pair of claims from different papers, a contradiction candidate score is computed using a pre-trained Natural Language Inference (NLI) model (cross-encoder/nli-deberta-v3-base). Claim pairs with embedding similarity above threshold θ_1 are evaluated by the NLI model, and pairs with a contradiction probability above threshold τ are forwarded to Stage 2.
Stage 2 — LLM-Based Confirmation and Classification: Each candidate pair is submitted to the LLM with a structured prompt instructing it to: (a) confirm whether the pair is genuinely contradictory, partially contradictory, or non-contradictory; (b) assign a conflict type from the four-dimensional taxonomy; and (c) generate an explanation and reconciliation statement.

Four-Dimensional Conflict Taxonomy. We define four orthogonal conflict types:

1. **Methodological Conflict:** Different conclusions due to different methods, metrics, or experimental setups.
2. **Domain-Specificity Conflict:** Claims conflict because they hold in different application domains.

3. **Temporal Conflict:** One finding has been superseded or qualified by more recent work.
4. **Definitional Conflict:** Apparent conflict arising from different definitions of the same term.

The Conflict Registry. For each confirmed contradiction, the Resolution Agent generates a *Reconciliation Statement*—a structured explanation presenting both claims, identifying the conflict type, and synthesizing a qualified assertion true under specified conditions. These are stored in a *Conflict Registry* and integrated into the review as a dedicated "Contested Claims and Methodological Disputes" section.

4.4 Formalized Research Gap Ontology (FRGO)

Gap Object Schema. Each research gap is represented as a structured typed object containing: `gap_type`, `topic_cluster`, `missing_intersection`, `gap_statement`, `evidence_papers`, `evidence_type`, `implied_by`, `confidence`, `temporal_position`, `gap_class`, and critically, a `falsifiability` statement describing what a paper that fills the gap would contain.

Gap Type Taxonomy. Gaps are classified into four types:

– **Unexplored Intersection:** A combination of established topics that no paper has addressed together, identified by finding topic cluster pairs sharing no papers in the DSKG.
– **Underexplored Area:** A topic addressed by fewer papers than its citation frequency would predict.
– **Contradictory State:** Fundamental disagreement with no resolution paper providing definitive synthesis.
– **Methodological Gap:** An established finding not reproduced using alternative methodologies.

Empirical Gap Validation Protocol. The defining feature of the FRGO is its empirical evaluability. Each gap object includes an explicit falsifiability statement. Evaluation uses a held-out set of papers published after the corpus cutoff date. *Gap Precision at K* (GP@K) is computed as the proportion of top-K identified gaps filled by papers in the held-out set.

4.5 Dynamic Scientific Knowledge Graph (DSKG)

Graph Schema. The DSKG is a directed heterogeneous graph $G = (V, E)$ implemented in NetworkX where:
Node Types V: Concept nodes (research concepts, datasets, metrics), Claim nodes (atomic scientific claims), Paper nodes (metadata), Method nodes (techniques, algorithms), and Finding nodes (empirical outcomes).
Edge Types E (Epistemic Relationships): *supports*, *contradicts* (bidirectional), *extends*, *qualifies*, *applies_ to*, *evaluated_ on*, *introduces*, and *outperforms*.

Graph-Based Synthesis Reasoning. The Synthesizer Agent generates the review by traversing the DSKG:

- **Thematic Clustering:** Louvain community detection identifies cohesive clusters representing natural review subsections.
- **Lineage Tracing:** Following *extends* and *introduces* edges reconstructs developmental history.
- **Contradiction Surfacing:** Traversing *contradicts* edges ensures comprehensive conflict coverage per thematic cluster.
- **Structural Gap Detection:** Disconnected subgraphs, sparse cross-cluster edges, and high-centrality nodes with few extensions indicate research gaps.

4.6 Temporal Knowledge Evolution Tracker (TKET)

The TKET models research threads as temporal sequences by associating DSKG nodes with publication timestamps. For each thread, it computes:

- **Consensus Trajectory:** How the dominant claim has shifted over time—positive (growing consensus), diverging (increasing dispute), or reversal (paradigm shift).
- **Method Emergence Curves:** First appearance and adoption rate of key methods within a thread.
- **Claim Qualification Patterns:** Instances where a claim is explicitly qualified, refined, or bounded by subsequent papers.

This enables *Diachronic Narrative* sections that describe how the field's understanding has evolved: "Early work from 2018–2020 established X. This was qualified in 2021–2022 by evidence that X holds only under condition Y. The current consensus as of 2024 is Z."

4.7 Multi-Agent Architecture

Agent Roster. Table 2 summarizes the eleven specialized agents in the framework.

The Autonomy Loop. The Autonomy Loop Controller evaluates the Critic Agent's quality assessment against a configurable threshold Q^*. The loop proceeds as specified in Algorithm 1.

5 System Architecture and Implementation

5.1 Layered Architecture

The system is organized into five layers as shown in Table 3.

Table 2. Agent roster and responsibilities.

Agent	Responsibility
Goal Interpreter	Refines user topic into precise query and subtopic decomposition
Planner	Generates task execution DAG with dependency ordering
Retriever	Orchestrates Semantic Scholar and arXiv for corpus construction
Claim Extractor	Extracts atomic claims from paper sections via structured prompts
Confidence Scorer	Implements CGCERS formula for confidence grading
Contradiction Detector	Two-stage CDRA with NLI filtering and LLM confirmation
Resolution Agent	Generates reconciliation statements for confirmed conflicts
DSKG Builder	Assembles papers, claims, and relations into the knowledge graph
Gap Analyzer	FRGO formalization using DSKG structural analysis
Temporal Tracker	Models diachronic trajectories and consensus evolution
Synthesizer	Graph-traversal-based review generation with confidence annotations
Critic	Evaluates review quality on five dimensions

5.2 Technology Stack

The implementation uses the following key technologies:

- **LLM Engine:** Google Gemini 2.0 Flash as primary reasoning engine via LangChain, with a modular `BaseAgent` class enabling seamless switching to OpenAI or Anthropic providers.
- **Knowledge Graph:** NetworkX for in-memory heterogeneous directed graph with JSON persistence, supporting Louvain community detection, PageRank and betweenness centrality, and lineage tracing.
- **Vector Store:** ChromaDB with cosine similarity for semantic claim search, supporting consensus computation and contradiction candidate identification.
- **PDF Processing:** GROBID (via Docker) for structured TEI XML parsing of academic PDFs with section segmentation and reference extraction; PyMuPDF as a heuristic fallback.
- **NLI Model:** Pre-trained cross-encoder/nli-deberta-v3-base for Stage 1 contradiction candidate filtering.
- **Embeddings:** Google Gemini embedding API (models/embedding-001) with sentence-transformers (all-MiniLM-L6-v2) as local fallback.
- **Frontend:** Streamlit with Plotly for evaluation radar charts and streamlit-agraph for interactive knowledge graph visualization.

Algorithm 1 Autonomy Loop Controller

```
Require: Topic T, quality threshold Q*, max iterations I_max
Initialize PipelineState, DSKG, ConflictRegistry, GapStore
state ← GoalInterpreter.run(T)
state ← Planner.run(state)
for i = 1 to I_max do
    state ← Retriever.run(state)
    state ← PDFProcessor.process(state)
    state ← ClaimExtractor.run(state)
    state ← ConfidenceScorer.run(state)
    state ← ContradictionDetector.run(state)
    state ← ResolutionAgent.run(state)
    state ← DSKGBuilder.run(state)
    state ← GapAnalyzer.run(state)
    state ← TemporalTracker.run(state)
    state ← Synthesizer.run(state)
    state ← Critic.run(state)
    if state.quality ≥ Q* then
        break
    else
        Apply targeted refinements based on Critic feedback
    end if
end for
return state.review
```

5.3 Claim Extraction Pipeline

The claim extraction pipeline processes each paper through the following stages:

1. **PDF Ingestion:** GROBID parses the PDF into TEI XML, extracting structured sections (abstract, introduction, methods, results, discussion, conclusion), tables, and figures. If GROBID is unavailable, PyMuPDF performs raw text extraction with heuristic section segmentation using regex-based header detection.
2. **Section-Level Extraction:** Each section is submitted to the Claim Extractor Agent with a structured prompt specifying the expected output schema (claim_text, claim_type, confidence_indicators, subject_entities, condition_qualifiers).
3. **Embedding Generation:** Each extracted claim receives a vector embedding via the Google Gemini embedding API for subsequent similarity-based operations.
4. **CGCERS Scoring:** The Confidence Scorer computes consensus, recency, and contradiction scores using Equation 1, querying the ChromaDB claim store for semantically similar claims from other papers.
5. **DSKG Insertion:** Scored claims are added as nodes to the DSKG with edges linking them to their source papers, related concepts, and methods.

Table 3. Five-layer system architecture with component mapping.

Layer	**Components**
Interface	Streamlit web interface with topic input, interactive KG viewer, Conflict Registry, Gap Explorer, Review output, and Evaluation Dashboard
Orchestration	Planner Agent, Autonomy Loop Controller, PipelineState management
Agent Execution	11 specialized agents with structured I/O schemas and shared message bus
Knowledge	DSKG (NetworkX), ClaimStore (ChromaDB), ConflictRegistry (JSON), GapStore (JSON)
External Integration	Semantic Scholar API, arXiv API, GROBID PDF processor, Google Gemini LLM API

5.4 Contradiction Detection Pipeline

The two-stage CDRA pipeline operates as follows:

1. **Candidate Identification:** For each claim, ChromaDB retrieves semantically similar claims from different papers (similarity $> \theta_1 - 0.2$). The NLI model computes contradiction probabilities for each candidate pair.
2. **Filtering:** Pairs with NLI contradiction score $> \tau$ (default $\tau = 0.6$) are forwarded to Stage 2. The top 50 candidates (sorted by NLI score) are processed to manage API costs.
3. **LLM Confirmation:** Each candidate pair, augmented with paper metadata (title, year), is submitted to the LLM for confirmation, classification, and explanation generation.
4. **Resolution:** Confirmed contradictions are passed to the Resolution Agent, which generates detailed reconciliation statements and stores the complete `ConflictObject` in the Conflict Registry.
5. **DSKG Update:** Bidirectional *contradicts* edges are added to the DSKG between confirmed conflicting claims.

5.5 Gap Analysis Pipeline

The FRGO gap analysis pipeline combines structural graph analysis with LLM-based formalization:

1. **Community Detection:** Louvain algorithm partitions the DSKG into thematic clusters.
2. **Structural Analysis:** Four types of structural gap indicators are computed:
   - Sparse cross-cluster edge density → Intersection Gaps
   - High PageRank centrality with few *extends* edges → Underexplored Gaps
   - Dense contradiction edges without resolution papers → Contradictory State Gaps

   – Methods applied to few domains → Methodological Gaps
3. **LLM Formalization:** Each structural indicator is submitted to the Gap Analyzer Agent for formalization into a complete FRGO gap object with falsifiability statement.
4. **Gap Store Insertion:** Formalized gaps are persisted to the JSON-backed Gap Store with confidence ranking.

5.6 Data Models

The system uses Pydantic-based data models ensuring type safety and validation. The central `PipelineState` model maintains the shared state across all agents, containing the paper corpus, extracted claims, conflict pairs, gap objects, temporal threads, the generated review, and evaluation results. Key model classes include `AtomicClaim`, `ConfidenceGradedClaim`, `ConflictPair`, `ConflictObject`, `GapObject`, `TemporalThread`, `GeneratedReview`, and `CriticEvaluation`.

5.7 Configuration Management

All system parameters are managed through a centralized Pydantic Settings class loading from a `.env` file, including: LLM provider selection and API keys, GROBID server URL, pipeline parameters (max papers, autonomy iterations, quality threshold, temporal scope), CGCERS weighting parameters (α, β, γ, λ), and contradiction detection thresholds (θ_1, τ).

6 Evaluation Strategy

6.1 Evaluation Overview

Evaluation of literature review systems is a notoriously difficult problem because there is no single ground truth review for a given topic. Our evaluation strategy addresses this through four complementary methods: automated metric computation against expert surveys, novel task-specific metrics, human expert evaluation, and the empirical gap validation protocol.

6.2 Benchmark Datasets and Domains

Three domains are selected representing different characteristics:

1. **Retrieval-Augmented Generation (RAG):** A rapidly evolving domain with frequent contradictions and active methodological debate.
2. **Graph Neural Networks (GNN):** A more mature domain with established consensus on core findings and well-defined evaluation benchmarks.
3. **Low-Resource Machine Translation:** A domain with significant methodological diversity and cross-lingual challenges.

For each domain, a corpus of 50–150 primary research papers spanning 2017–2025 is assembled via Semantic Scholar and arXiv. Gold standard expert-authored surveys are withheld from the retrieval corpus and used only for evaluation. A held-out set of papers published in 2025–2026 is assembled for gap validation.

6.3 Novel Task-Specific Metrics

We introduce five evaluation metrics designed to assess the specific capabilities of uncertainty-aware synthesis:

Claim Coverage Score (CCS): Measures the proportion of key claims identified in the gold standard survey that are present (in semantically equivalent form) in the generated review. Computed using embedding similarity with threshold matching. CCS measures content recall from an expert perspective.

Confidence Calibration Score (CalibScore): Measures whether confidence assignments are calibrated. For claims marked as High Confidence, the proportion that experts agree are well-established should be high. Calibration curves are plotted for each confidence category to assess alignment between system scores and expert consensus.

Contradiction Detection F1 (CD-F1): A held-out contradiction pair set is constructed by domain experts identifying genuine contradictions within each corpus. Precision is the proportion of system-identified contradictions confirmed by experts; recall is the proportion of expert-identified contradictions found by the system.

Gap Precision at K (GP@K): The top-K gap objects generated by the system are evaluated against the held-out future paper set. A gap is considered validated if a paper in the held-out set directly addresses the gap as specified by the falsifiability statement. GP@5 and GP@10 are reported.

Temporal Coherence Score (TCS): Expert reviewers assess whether the temporal narrative produced by the TKET accurately represents the historical development of the field, rated on a 5-point Likert scale.

6.4 Human Evaluation Protocol

Human evaluation is conducted with five domain experts per domain. Reviewers are presented with the generated review alongside the gold standard survey and baseline system outputs. Evaluation dimensions include:

- **Factual Accuracy:** Are claims accurately attributed and correctly stated?
- **Synthesis Quality:** Does the review integrate multiple papers rather than merely summarizing individually?
- **Contradiction Handling:** Are conflicting findings acknowledged and addressed?
- **Gap Usefulness:** Are identified research gaps specific and actionable?

– **Temporal Awareness:** Does the review accurately convey the historical development of the field?

Each dimension is rated on a 7-point Likert scale. Inter-rater reliability is measured using Cohen's Kappa. Reviewers are blind to which system generated which output.

6.5 Ablation Study Design

To validate the contribution of each component, a systematic ablation study compares the following configurations (Table 4):

Table 4. Ablation study configurations. ✓ indicates component inclusion.

Configuration	CGCERS	CDRA	FRGO	DSKG	TKET
Full System	✓	✓	✓	✓	✓
−CGCERS		✓	✓	✓	✓
−CDRA	✓		✓	✓	✓
−FRGO	✓	✓		✓	✓
−DSKG (flat retrieval)	✓	✓	✓		✓
−TKET	✓	✓	✓	✓	
Baseline (RAG-only)					

7 Limitations

Despite the contributions of this work, several limitations should be acknowledged:

1. **LLM Hallucination Risk in Claim Extraction:** Despite structured prompting, LLMs may hallucinate claim details not present in the source paper. This is mitigated through post-extraction verification that checks extracted claims against source text using substring and semantic matching, but cannot be fully eliminated.
2. **Knowledge Graph Construction Accuracy:** Relationship classification errors during DSKG construction may introduce incorrect edges that propagate through synthesis. The Critic Agent's coverage evaluation partially mitigates this but cannot catch all errors.
3. **Scalability:** The contradiction detection pairwise comparison has quadratic complexity $O(|\mathcal{C}|^2)$ in the number of claims. For corpora exceeding 500 papers, approximate nearest-neighbor search is used for candidate filtering, which may reduce recall.
4. **Evaluation Ground Truth Limitations:** Expert-authored survey papers used as gold standards may themselves contain biases, omissions, and errors, limiting their validity as absolute ground truth.

5. **Domain Generalization:** The pre-trained NLI model is primarily trained on NLP and computer science text. Performance on domains with different rhetorical conventions (e.g., medicine, social sciences) may require domain-specific fine-tuning.
6. **PDF Processing Quality:** GROBID's section segmentation accuracy varies with PDF formatting quality, particularly for older papers without standardized layouts.

8 Future Work

Several directions for future work are envisioned:

- **Citation Intent Classification:** Incorporating citation intent classification (supportive, contrasting, neutral) from tools such as SciCite directly into DSKG edge construction for more accurate relationship modeling.
- **Domain-Specific Fine-Tuning:** Fine-tuning the Claim Extractor and Contradiction Detector agents on domain-specific corpora for medicine, law, and social sciences.
- **Real-Time Monitoring:** Extending the system to monitor newly published papers and automatically update the DSKG, Conflict Registry, and Gap Set when new work resolves a gap or introduces a new contradiction.
- **Peer Reviewer Simulation:** Adding a Peer Reviewer Agent that evaluates the generated review against venue-specific standards, simulating the review process for target publication venues.
- **Multi-Language Support:** Extending retrieval and processing to non-English language papers, important for fields with significant non-English publication traditions.
- **Interactive Refinement:** Enabling users to provide real-time feedback on specific claims, contradictions, or gaps through the Streamlit interface, creating a human-in-the-loop refinement cycle.

9 Conclusion

This paper has presented a novel autonomous multi-agent framework for scientific literature review generation that addresses three fundamental limitations of existing systems: the absence of epistemic grounding for synthesized claims, the inability to detect and resolve contradictions across papers, and the generation of vague, unverifiable research gap statements.

The system's five core technical contributions—Confidence-Graded Claim Extraction and Reliability Scoring (CGCERS), the Contradiction Detection and Resolution Agent (CDRA), the Formalized Research Gap Ontology (FRGO), the Dynamic Scientific Knowledge Graph (DSKG), and the Temporal Knowledge Evolution Tracker (TKET)—collectively advance the state of the art in automated scientific reasoning. The CGCERS formula (Equation 1) provides calibrated confidence scores based on multi-paper consensus, temporal recency, and counter-evidence. The CDRA's two-stage architecture combining NLI model filtering with LLM confirmation enables efficient yet accurate contradiction detection with a structured four-dimensional taxonomy. The FRGO introduces empirically evaluable gap objects with explicit falsifiability statements—a first in the automated survey generation literature.

The proposed evaluation framework, introducing CCS, CD-F1, GP@K, TCS, and CalibScore, addresses the longstanding evaluation inadequacy in automated literature review research and provides a reusable assessment protocol for future work. The complete system has been implemented using Google Gemini as the primary LLM engine with a modular architecture supporting provider switching, NetworkX for graph-based reasoning, ChromaDB for semantic search, and GROBID for structured PDF processing, all orchestrated by an autonomy loop that iteratively refines outputs based on critic feedback.

By moving beyond static pipeline summarization toward a graph-centric, epistemically grounded, contradiction-aware, and temporally sensitive synthesis architecture, this work takes a meaningful step toward AI systems capable of genuine scientific reasoning partnership—not merely retrieving and rewording existing text, but reasoning about the structure of knowledge itself.

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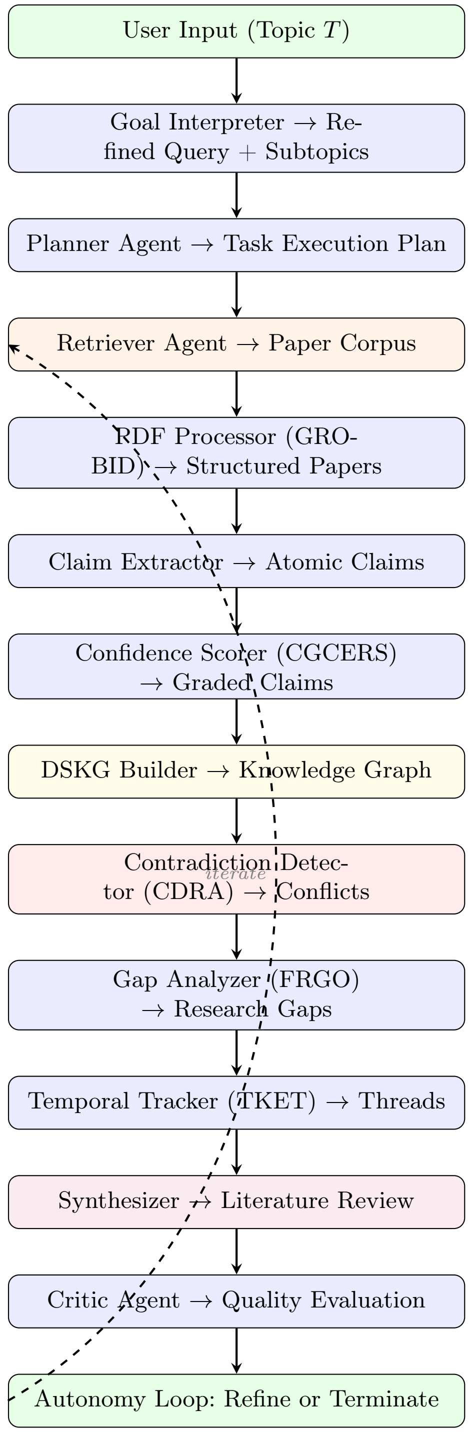

Fig. 1. High-level execution flow of the multi-agent autonomy loop. Dashed arrow indicates iterative refinement triggered by the Critic Agent's quality assessment.